

Kernel Geometry Governs Robustness under Distribution Shift: A Controlled Multi-Dataset Study of Quantum and Classical Kernels

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Abstract

Whether quantum kernels improve robustness under distribution shift is usually asked as a quantum-versus-classical question. We argue, and show empirically, that it is more productively asked as a question about kernel geometry. We present a controlled kernel-swap study in which the classifier, preprocessing, split construction, and family-internal model-selection logic are held fixed while only the kernel varies, instantiated on four security benchmarks spanning two modalities and three drift mechanisms (EMBER static malware; UNSW-NB15 and ToN-IoT network flows under synthetic covariate shift and natural regime drift), two classifier families (support vector classification and a Laplace-approximation Gaussian process classifier with calibrated uncertainty), and 72 principal settings with 15 repeated runs each. Against the customary linear and RBF baselines, fidelity-based quantum kernels improve out-of-distribution balanced accuracy nearly uniformly (69–71 of 72 settings; Wilcoxon $p < 10^{-9}$ on network data). Against a broader classical family, a median-heuristic Laplacian kernel neutralizes the advantage on EMBER—revealing that the operative property is not quantumness but geometry—yet when both families receive the same bandwidth-tuning freedom the quantum advantage re-emerges on EMBER and remains dominant under natural network drift (54 of 54 settings under the probabilistic

classifier, $p = 1.6 \times 10^{-10}$). A mechanistic analysis explains these outcomes: within each setting, out-of-distribution accuracy tracks the survival of centered kernel-target alignment under shift, a regularity that holds in 89–99% of 360 setting-seed units across all four datasets, with high effective rank of the training kernel as its structural precondition. The classical-favorable exceptions of the original comparison are precisely the settings where the selected quantum kernel collapses to low-rank, classical-like geometry. Beyond the specific quantum-versus-classical verdict, the study contributes a reusable, fully reproducible protocol for attributing robustness effects to kernel geometry, and a design principle—prefer kernels whose label-aligned geometry survives shift—that spans classical and quantum families alike.

Keywords: quantum machine learning; quantum kernels; distribution shift; out-of-distribution robustness; kernel methods; kernel-target alignment; Gaussian processes; malware detection; network intrusion detection

1 Introduction

Quantum kernel methods are one of the clearest and most actively studied directions in quantum machine learning. By embedding classical data into quantum states and defining similarity through state overlaps, they provide a principled way to study how quantum-induced feature geometry affects supervised learning while keeping the downstream classifier fixed [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). At the same time, the recent QML literature has made an equally important point: empirical conclusions about quantum models can depend strongly on the data regime, the encoding, and the evaluation protocol [Huang et al. \(2021\)](#); [Schuld et al. \(2021\)](#); [Gil-Fuster et al. \(2024\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#). For that reason, claims about the usefulness of quantum kernels are most informative when they are established under controlled, reproducible, and carefully matched comparisons rather than through heterogeneous pipelines or loosely aligned baselines.

This need for control becomes even more important under distribution shift. In many applied settings, the relevant question is not only whether a model separates classes well under in-distribution (ID) conditions, but whether its behavior remains

reliable once the evaluation distribution changes. The broader distribution-shift and domain-generalization literature has repeatedly shown that apparent robustness gains can weaken, disappear, or become difficult to interpret when model-selection strategy, hyperparameter search, split construction, or evaluation standardization are not carefully controlled [Quiñonero-Candela et al. \(2008\)](#); [Zhou et al. \(2023\)](#); [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#). A convincing robustness study must therefore satisfy two requirements at once: it must isolate the factor of interest tightly enough to support attribution, and it must evaluate that factor broadly enough to determine whether the observed behavior reflects a meaningful regime-level effect rather than a narrow configuration artifact.

These issues are especially acute in security applications. Drift is not incidental to the problem; it is part of the problem definition itself. Static PE malware classification has been widely studied through benchmarks such as EMBER [Anderson and Roth \(2018\)](#), while more recent work has highlighted the gap between laboratory evaluation and performance on drifted malware, as well as the difficulty of adapting with limited post-drift data [Barbero et al. \(2022\)](#); [Li et al. \(2025\)](#). The emergence of updated resources such as EMBER2024 further reflects an active push toward more realistic, broader, and more deployment-relevant malware evaluation [Joyce et al. \(2025\)](#). Network intrusion detection faces the same structural challenge: traffic distributions move as services, infrastructure, and attack campaigns evolve, and public benchmarks such as UNSW-NB15 and ToN-IoT expose exactly this kind of regime change [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#). Taken together, this literature suggests that a useful security-ML study should not merely report accuracy on a convenient split; it should examine how model choices behave when the evaluation explicitly targets shift, and it should do so across more than one data modality and drift mechanism.

Kernel methods are a natural setting in which to ask such questions cleanly. Because kernel-based support vector machines separate the choice of classifier from the

choice of similarity function, they offer a principled way to study how representational geometry affects ID and OOD behavior while keeping the learning machinery fixed [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#). This perspective becomes particularly interesting in quantum machine learning, where quantum kernels are defined through implicit feature maps into quantum Hilbert spaces and estimated via state overlaps or related fidelity-based similarities [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). From this viewpoint, quantum kernels are not a vague alternative model class, but a concrete family of kernel constructions whose empirical value can—and should—be tested under the same standards applied to classical kernels.

At the same time, the quantum-kernel literature has matured to the point where optimistic demonstrations and critical scrutiny must now be read together. Foundational work established the feature-space interpretation of quantum kernels and motivated them as expressive alternatives to classical similarities [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#). Subsequent studies emphasized that performance in quantum machine learning can depend strongly on the data regime, the encoding, and the effective geometry induced by the model [Huang et al. \(2021\)](#); [Schuld et al. \(2021\)](#). More recent work has sharpened this message further: understanding generalization in quantum machine learning requires care beyond classical intuition, and benchmarking conclusions can be highly sensitive to protocol choices [Gil-Fuster et al. \(2024\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#). This makes broad claims of superiority difficult to defend and reinforces the need for conservative, protocol-aware empirical studies.

The gap we address lies precisely at this intersection, and it has three parts. First, there remains limited evidence from controlled, variability-aware comparisons between classical and quantum kernel families in security-relevant settings where out-of-distribution robustness is part of the evaluation target: existing studies often differ

simultaneously in model class, preprocessing, search space, or data protocol, making it difficult to attribute observed gains to the kernel itself. Second, when such comparisons are attempted, the classical reference set is typically narrow—linear and RBF kernels under a single classifier—so an apparent quantum advantage may reflect the weakness of the baselines rather than a property of the quantum family; probabilistic classifiers with principled uncertainty quantification, and classical kernels with richer spectral structure, are usually absent. Third, and most fundamentally, empirical wins and losses are rarely accompanied by an account of *why* a kernel family should behave better or worse under shift, leaving results as observations in search of an explanation. For a question as specific as whether quantum kernels help under distribution shift, these confounds and omissions are not peripheral; they are decisive.

In this paper, we study that question through a strictly controlled *kernel-swap* protocol, instantiated broadly enough to close all three gaps at once. Within each experimental setting we hold the classifier, the preprocessing pipeline, the split definitions, and the family-internal model-selection logic fixed, and vary only the kernel. The protocol is applied to four security benchmarks spanning two data modalities and three drift mechanisms: EMBER static PE malware under two synthetic covariate-shift constructions [Anderson and Roth \(2018\)](#), and UNSW-NB15 and ToN-IoT network-flow scenarios under both an analogous synthetic construction and *natural* regime drift, in which the out-of-distribution pool consists of genuinely drifted traffic [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#). Two classifier families consume the same precomputed kernels: support vector classification, and a binary Laplace-approximation Gaussian process classifier [Rasmussen and Williams \(2006\)](#) that adds calibrated predictive probabilities—and hence principled uncertainty quantification under shift—without changing anything else in the pipeline. The classical family ranges from the customary linear and RBF baselines to polynomial, Laplacian, and Matérn kernels with train-only median-heuristic length scales; the quantum family

comprises fidelity kernels induced by commonly used feature-map families. Crucially, both families receive the same bandwidth-tuning freedom (a gamma sweep for RBF, an angle-scaling sweep for the quantum maps), so that neither side is compared at a fixed operating point the other is allowed to tune away from [Shaydulin and Wild \(2022\)](#); [Canatar et al. \(2023\)](#). The main evaluation comprises 72 principal settings (18 on EMBER, 54 on network flows), each repeated across 15 runs, with paired Wilcoxon signed-rank tests and Holm correction on all family-level comparisons.

Beyond measuring who wins, we ask why. For every kernel, dimension, and setting we compute geometry descriptors of the induced Gram matrices—the effective rank of the training kernel [Roy and Vetterli \(2007\)](#), centered kernel-target alignment on the ID and OOD splits [Cristianini et al. \(2001\)](#); [Cortes et al. \(2012\)](#), kernel concentration statistics, and the geometric difference between classical and quantum kernels [Huang et al. \(2021\)](#)—and test which of these quantities predict out-of-distribution behavior within and across settings. This mechanistic layer converts the benchmark from a scoreboard into an instrument for attribution.

The resulting picture is sharper than a quantum-versus-classical verdict, and more useful. Against linear and RBF baselines, fidelity-based quantum kernels improve OOD balanced accuracy nearly uniformly across all four datasets (69–71 of 72 settings under both classifiers). Against the extended classical family, the advantage on EMBER is neutralized by a median-heuristic Laplacian kernel—demonstrating that the operative property is geometric rather than quantum—but re-emerges under symmetric bandwidth tuning, and remains dominant on network flows, where under the probabilistic classifier the quantum family wins all 54 settings with gains exceeding run-to-run variability in the majority. In-distribution separability favors the quantum family in every EMBER setting regardless of the classical reference set. The mechanistic analysis explains all of these outcomes with one regularity: within a setting, OOD accuracy tracks the survival of centered kernel-target alignment under shift—positive in 89–99%

of 360 setting-seed units across all four datasets and both classifiers—with high effective rank of the training kernel as its structural precondition. The classical-favorable exceptions of the baseline comparison are exactly the settings where the selected quantum kernel collapses to a low-rank, classical-like geometry, and the Laplacian is exactly the classical kernel whose geometry most resembles the quantum maps.

Our contributions are fourfold. First, a controlled, reproducible kernel-swap protocol for robustness studies that extends across classifier families (including a probabilistic classifier with calibrated uncertainty), across a broad classical reference set with symmetric bandwidth tuning, and across datasets, modalities, and drift mechanisms. Second, a variability-aware regime map of quantum-versus-classical kernel behavior under shift on 72 settings with formal significance testing, including the honest negative result that a well-chosen classical kernel matches the quantum family on one modality at fixed bandwidth. Third, a mechanistic account—alignment survival under shift, enabled by high effective rank—that predicts within-setting OOD performance across every dataset we study, explains the structured exceptions, and yields the checkable prediction (which we confirm) that the classical kernels sharing this geometry are the competitive ones. Fourth, practical guidance and a fully public artifact: for OOD-sensitive kernel selection, prefer kernels whose label-aligned geometry survives shift; fidelity-based quantum maps exhibit this property natively, and on drifting network traffic they currently exploit it best.

2 Related Work

2.1 Distribution shift, domain generalization, and evaluation rigor

The broader literature on distribution shift has established that predictive performance can degrade substantially when training and test data are drawn from different

distributions, and that robust evaluation under shift requires more than reporting strong in-distribution performance [Quiñonero-Candela et al. \(2008\)](#); [Shimodaira \(2000\)](#); [Sugiyama et al. \(2007\)](#). Within machine learning, domain generalization (DG) has emerged as one of the main formal settings for studying out-of-distribution (OOD) performance without access to target-domain labels during training [Zhou et al. \(2023\)](#). This literature has also made clear that empirical conclusions about robustness are highly sensitive to evaluation design.

A particularly important development in this regard is the move toward benchmark-driven scrutiny of robustness claims. Gulrajani and Lopez-Paz showed in DomainBed that many reported improvements in DG are difficult to interpret because they depend strongly on inconsistencies in datasets, architectures, hyperparameter budgets, and model-selection criteria [Gulrajani and Lopez-Paz \(2021\)](#). WILDS extended this concern to real-world shifts by curating datasets whose train–test differences arise naturally across institutions, locations, and acquisition conditions, and showed that standard training can leave large OOD gaps even on modern pipelines [Koh et al. \(2021\)](#). Wild-Time further emphasized the importance of temporal shift as a distinct and practically relevant regime, highlighting that even when timestamps are available and temporal structure can in principle be exploited, existing methods still struggle to close the ID–OOD gap [Yao et al. \(2022\)](#).

These works are directly relevant to the present study, not because malware detection is a standard DG benchmark, but because they establish a methodological principle: if the goal is to compare robustness under shift, then fairness of protocol is part of the contribution, not a secondary implementation detail. Our work adopts this perspective by focusing on a comparison in which the classifier family, preprocessing, and split definitions are fixed, so that the role of the kernel can be interpreted with minimal confounding.

2.2 Malware detection under drift and evolving evaluation benchmarks

In malware detection, the relevance of distribution shift is especially acute. Malware ecosystems evolve continuously as attackers modify payloads, packers, obfuscation strategies, delivery mechanisms, and development toolchains. Static detection models are therefore exposed to changing distributions in ways that are operationally consequential rather than merely academic. Public benchmarks have played an important role in making this problem more reproducible. EMBER remains one of the most widely used open datasets for static PE malware detection and helped standardize large-scale evaluation in this area [Anderson and Roth \(2018\)](#). More recent resources such as EMBER2024 push this effort further by broadening scope, increasing scale, and enabling more holistic evaluation across multiple malware-related tasks [Joyce et al. \(2025\)](#). Other datasets such as BODMAS and SOREL-20M likewise reflect the field’s effort to move toward larger and more realistic benchmark settings [Yang et al. \(2021\)](#); [Harang and Rudd \(2020\)](#).

At the same time, a growing body of work has shown that strong laboratory performance does not guarantee robustness to evolving malware. Barbero et al. revisited malware classification under concept drift and showed that deployment-oriented settings raise challenges that are obscured by simpler static evaluations [Barbero et al. \(2022\)](#). More recently, Li et al. studied adaptation to real drifted malware under limited post-drift supervision, reinforcing the point that temporal degradation is not hypothetical and that realistic recovery from drift may be data-constrained [Li et al. \(2025\)](#). Taken together, these studies support a clear conclusion: in malware detection, robustness under shift is not a niche desideratum but a primary evaluation target.

Our work differs from this line in emphasis. Rather than proposing a new drift-adaptation mechanism, drift detector, or continual-update strategy, we ask a narrower but complementary question: under a tightly controlled protocol, does the choice of

kernel family itself materially affect OOD robustness? This question is well motivated by the malware literature, yet comparatively underexplored in a way that isolates kernels from the many other moving parts of the pipeline.

2.3 Kernel methods, quantum kernels, and careful benchmarking in QML

Kernel methods provide a natural framework for studying robustness under controlled model comparisons because they separate the learning algorithm from the similarity function used to compare inputs. In classical machine learning, support vector machines and related kernel methods have long provided a principled route for controlling model capacity while varying the geometry of the induced feature space [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#). This makes kernels especially attractive when the scientific question concerns representation and similarity rather than wholesale changes in classifier architecture.

Quantum kernel methods arise from the observation that data can be embedded into quantum states and compared through overlaps in a quantum feature space. Havlíček et al. popularized this perspective in the context of supervised classification with quantum-enhanced feature spaces [Havlíček et al. \(2019\)](#), while Schuld and Killoran provided a broader theoretical account linking quantum models to kernel methods through feature Hilbert spaces [Schuld and Killoran \(2019\)](#). Later work sharpened this connection by arguing that supervised quantum machine learning models can often be interpreted directly as kernel methods, making the kernel viewpoint central rather than incidental [Schuld \(2021\)](#). Additional work on data encoding further showed that representational behavior depends critically on how classical inputs are mapped into quantum states, which is directly relevant when comparing quantum feature maps as alternative kernel constructions [Schuld et al. \(2021\)](#).

However, the contemporary QML literature also makes clear that promising empirical behavior should not be read uncritically. Huang et al. showed that the role of data can be decisive in quantum machine learning, suggesting that observed performance gains are deeply regime-dependent rather than model-intrinsic [Huang et al. \(2021\)](#). Caro et al. studied generalization from few training examples, while Gil-Fuster et al. later argued that understanding generalization in QML requires rethinking familiar classical intuitions, since standard explanations may fail to account for observed behavior in quantum models [Caro et al. \(2022\)](#); [Gil-Fuster et al. \(2024\)](#). In parallel, broader perspective pieces have emphasized both the promise and the unresolved challenges of QML as a field [Cerezo et al. \(2022\)](#).

Most importantly for the present work, recent benchmarking studies have raised the standard for empirical claims in QML. Bowles et al. argued that experimental design choices can dominate conclusions and showed, in a broad simulated benchmark, that out-of-the-box classical models often remain strong baselines against quantum alternatives [Bowles et al. \(2024\)](#). Schnabel and Roth extended this benchmarking perspective specifically to quantum kernel methods, conducting a large-scale study across many design choices and showing that performance depends strongly on kernel type, encoding strategy, and hyperparameter configuration rather than on a simple classical-versus-quantum dichotomy [Schnabel and Roth \(2025\)](#). Related controlled evidence from adjacent QML settings points in the same methodological direction: recent work in hybrid quantum reinforcement learning shows that varying multiple pipeline components at once can obscure attribution, and highlights the importance of component-isolating, reproducible evaluation under fixed experimental conditions [Lazaro et al. \(2025\)](#). This line of work is especially relevant to our paper because it supports a conservative framing: the value of quantum kernels, if any, must be demonstrated under controlled and transparent comparisons, and should be expected to be regime-dependent.

Our study is positioned at the intersection of these three literatures. From the distribution-shift literature, we adopt the principle that robustness claims must rest on evaluation protocols that control major confounds. From the malware literature, we take the practical motivation that temporal and distributional change are intrinsic to realistic security deployment. From the QML literature, we take both the motivation for studying quantum kernels and the caution that benchmarking quality is central to interpretation. Relative to prior work, our contribution is not to claim universal superiority for quantum kernels, nor to propose a new adaptation method for drifted malware, but to provide a controlled, variability-aware comparison of classical and quantum kernel families in a malware-detection setting where OOD robustness is part of the primary target of evaluation.

3 Methods

3.1 Problem setting

We study binary security classification—malware versus benign software, and attack versus benign traffic—under distribution shift. Let D_{ID} denote the in-distribution data-generating process and D_{OOD} an out-of-distribution process induced by a shift mechanism, which may be synthetically constructed or naturally occurring. Given a training set $T \sim D_{\text{ID}}$, we evaluate the learned classifier on both an ID test set S_{ID} and an OOD test set S_{OOD} . Our objective is not only to measure separability under the i.i.d. assumption, but to compare how kernel families behave when generalization is evaluated explicitly under shift.

The central methodological requirement of the study is attribution. Because the scientific question concerns the effect of the kernel, comparisons must avoid confounds from changes in classifier family, preprocessing, optimization protocol, or split design. We therefore adopt a strictly controlled *kernel-swap* protocol in which all components of the pipeline are held fixed except the kernel function itself. This design follows

the same evaluation logic that motivates rigorous OOD benchmarking in the broader distribution-shift literature: when the target of inference is robustness, protocol control is not an implementation detail but part of the contribution.

3.2 Datasets and split construction

The protocol is instantiated on four benchmark scenarios spanning two data modalities. The first is EMBER, the standard public benchmark for static PE malware detection [Anderson and Roth \(2018\)](#), which is widely used, supports reproducible feature-based experimentation, and has motivated later benchmark extensions such as EMBER2024 [Joyce et al. \(2025\)](#). The remaining three are network-flow intrusion scenarios: two derived from UNSW-NB15 (with DoS and Reconnaissance, respectively, as the drifting attack category) [Moustafa and Slay \(2015\)](#) and one from ToN-IoT (Scanning) [Alsaedi et al. \(2020\)](#). In all cases our goal is not to benchmark absolute detection performance on a standard split, but to construct controlled ID/OOD regimes that allow kernel-family comparisons under distribution shift.

For each experimental setting, we adopt a fixed three-way split design:

$$T, \quad S_{\text{ID}}, \quad S_{\text{OOD}},$$

where T is used for training, S_{ID} measures in-distribution performance, and S_{OOD} measures robustness under the induced shift. All models compared within a setting use exactly the same split definitions. This point is essential: a classical and a quantum model are never compared on different train/test partitions.

EMBER shift constructions.

On EMBER we operationalize shift through two controlled OOD-construction variants, denoted $m1$ and $m2$, both built on the histogram + byte-entropy feature blocks and both designed to preserve class balance across splits.

Variant m1: sparsity-score extremes within class.

For each sample, we compute a sparsity-related score on a predefined feature block. In the main configuration used in the experiments, this score corresponds to the number of non-zero entries within the selected block. The OOD set is formed by taking, within each class separately, the low-score and high-score extremes, i.e., the two tails of the within-class score distribution. The remaining samples form an ID pool from which the training and ID test sets are obtained by stratified sampling. Intuitively, this construction creates an OOD regime driven by atypical sparsity structure while keeping the label distribution under control.

Variant m2: distance-to-train extremes within class.

We first fix the final training set at an exact target size. Using only this training set, we compute class-conditional centroids and assign each non-training sample a score equal to its Euclidean distance to the centroid of its own class. To reduce noise and avoid leakage, this distance can be computed in a TruncatedSVD-projected space fitted on the training set only. The OOD set is then defined, again within each class separately, by the low- and high-distance extremes from the non-training pool, with exact target OOD size and proportional class allocation. The ID test set is drawn from the remainder. This produces a complementary shift regime in which OOD samples are unusual relative to the final training geometry rather than to a sparsity statistic alone.

Network-flow scenarios and shift constructions.

Each network scenario is defined by a *reference* pool (traffic from the pre-drift regime) and a *current* pool (traffic containing the drifted attack campaign), prepared with a numeric-only feature policy that removes identifiers, label-derived columns, and categorical protocol fields; UNSW-NB15 scenarios retain 39 flow features and ToN-IoT 89. We verified explicitly that no feature is label-derived: the maximum feature-label

correlation across scenarios is 0.62 (attained by well-known legitimate features such as `sttl`), far from the near-perfect correlation that would signal leakage, and the verification script is part of the public artifact. On these pools we instantiate two shift mechanisms. The first, *m2-centroid*, is the direct analog of EMBER’s *m2*: within the class-balanced reference pool, a provisional training pool is fixed, class-conditional centroids are computed in a train-only MaxAbs+TruncatedSVD space, and the OOD pool consists of the within-class low- and high-distance tails. The second, *natural drift*, uses no synthetic construction at all: the training and ID pools are drawn from the reference regime and the OOD pool from the class-balanced current regime, so the evaluation measures robustness to genuinely drifted traffic rather than to constructed covariate extremes.

Together, these constructions are not intended as universal models of real-world drift. Rather, they provide reproducible shift mechanisms of three distinct kinds—atypical-sparsity extremes, train-geometry extremes, and natural regime drift—that are sufficiently different to support regime-level conclusions while preserving comparability across kernel families. In every setting, all models are compared on exactly the same T , S_{ID} , and S_{OOD} .

3.3 Controlled kernel-swap protocol

Our core experimental principle is to isolate the kernel as the only variable of interest. Accordingly, all compared models share the following components:

- **Classifier:** within each comparison, a single classifier consuming precomputed Gram matrices. Two classifier families are studied in parallel: Support Vector Classification with `kernel="precomputed"`, and a binary Gaussian process classifier under the Laplace approximation.
- **Preprocessing:** the same scaling and dimensionality-reduction pipeline for all classical and quantum models.

- **Split design:** identical T , S_{ID} , and S_{OOD} within each setting.
- **Selection logic:** the same family-internal model-selection principle applied consistently to classical and quantum candidates.

This yields a comparison between kernel families rather than between heterogeneous model pipelines. In particular, it prevents observed OOD differences from being attributed to changes in classifier architecture, feature engineering, or split construction. The choice of precomputed-kernel classifiers is deliberate: it allows both classical and quantum similarities to be consumed by exactly the same decision rule, thereby making the comparison interpretable in kernel-method terms [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#).

The Gaussian process classifier implements the standard binary Laplace approximation with a logistic link (Algorithms 3.1 and 3.2 in [Rasmussen and Williams \(2006\)](#)), applied directly to the same unit-diagonal precomputed kernels as the SVC. Its role in the study is twofold. Methodologically, it tests whether the kernel-family conclusions are classifier-specific or transfer across learning machinery. Substantively, it produces calibrated predictive probabilities, which allows robustness under shift to be examined through the lens of uncertainty quantification—log-loss, Brier score, expected calibration error, and predictive entropy on ID versus OOD data—rather than through accuracy alone. Kernel-swap comparisons are always performed within a classifier, never across classifiers.

Although the classical and quantum candidate sets are not identical in cardinality, fairness in this study is defined at the protocol level: both families are evaluated under the same classifier, preprocessing, splits, dimensionality sweep, and family-internal selection criterion, so that the effect attributed to the kernel is not confounded by changes elsewhere in the pipeline.

3.4 Preprocessing and feature representation

All models use the same preprocessing pipeline. Let $x \in \mathbb{R}^p$ denote the original feature vector of the dataset at hand. We apply the following steps:

1. **MaxAbs scaling.** We first apply `MaxAbsScaler` to normalize feature magnitudes while preserving sparsity structure.
2. **Dimensionality reduction.** We project the scaled features using `TruncatedSVD` to a reduced representation of dimension

$$d \in \{4, 6, 8, 10, 12\}.$$

3. **Angle mapping.** For the quantum pipeline, the reduced coordinates are linearly mapped to an angle range $[0, \pi]$ before being passed to the quantum feature maps.

This shared preprocessing serves two purposes. First, it keeps the comparison fair by ensuring that classical and quantum kernels operate on the same reduced representation. Second, it makes representation size itself a controlled factor of the study, enabling both aggregated setting-level comparisons and dimension-level analyses. The dimensionality parameter d is therefore not a tuning artifact hidden in preprocessing, but an explicit axis of the experimental grid.

3.5 Kernel families

Baseline classical kernels.

The customary reference set consists of two standard baselines:

$$K_{\text{lin}}(x, x') = x^\top x', \tag{1}$$

$$K_{\text{rbf}}(x, x') = \exp\left(-\gamma \|x - x'\|_2^2\right), \tag{2}$$

with the RBF kernel evaluated using the standard `scale` convention for γ and the linear kernel cosine-normalized to unit diagonal. These are the baselines against which quantum kernels are most commonly compared, and they define our *classical-original* family.

Extended classical family.

Because an apparent quantum advantage over two baselines may say more about the baselines than about the quantum family, we additionally evaluate a broader *classical-extended* family: polynomial kernels of degree 2 and 3 (cosine-normalized), the Laplacian kernel $K(x, x') = \exp(-\|x - x'\|_1/\ell_1)$, and Matérn kernels with $\nu \in \{3/2, 5/2\}$, where all length scales ℓ are set by the median heuristic—the median pairwise distance on the training set—so that every hyperparameter is determined from training data alone, without access to ID or OOD test information. These kernels were chosen deliberately to span a range of spectral behaviors: the Laplacian and low- ν Matérn kernels are less smooth than the RBF and induce Gram matrices with substantially heavier eigenvalue tails, which, as the mechanistic analysis will show, is precisely the property at stake under shift.

Bandwidth symmetry.

Kernel bandwidth is known to be a decisive hyperparameter for quantum kernels: rescaling the input encoding plays the same role for fidelity kernels that γ plays for the RBF kernel, and can qualitatively change generalization [Shaydulin and Wild \(2022\)](#); [Canatar et al. \(2023\)](#). A comparison in which the classical family may tune its bandwidth while the quantum family is pinned to a fixed encoding scale (or vice versa) is therefore not protocol-fair. In the bandwidth-symmetric portion of the study, both families receive the same tuning axis: the RBF kernel is evaluated at $\gamma = f \cdot \gamma_{\text{scale}}$ for $f \in \{0.1, 0.3, 1, 3, 10\}$, and every quantum feature map is evaluated at angle scalings

$\{0.5, 1, 2\}$ of the canonical $[0, \pi]$ encoding. Family-internal selection then operates over the enlarged candidate sets exactly as before.

Quantum kernels.

Quantum kernels are defined through an implicit feature map $\phi_Q(\cdot)$ induced by a parameterized quantum circuit, with similarity measured through fidelity-like overlaps [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). For two inputs x and x' , the corresponding kernel entry can be written abstractly as

$$K_Q(x, x') = |\langle \phi_Q(x), \phi_Q(x') \rangle|^2,$$

or equivalently through the overlap of states prepared by input-dependent quantum feature maps.

In the experiments, we evaluate multiple commonly used fidelity-based feature-map families chosen to represent distinct inductive biases: ZZ-type maps, PauliXZ-type maps, and Z-map variants. Concretely, the candidate set includes two ZZ configurations with different repetition depths, one PauliXZ configuration, and one Z-map configuration. We emphasize that the study is not attempting to exhaust the quantum-kernel design space; rather, it compares a representative set of commonly used fidelity-based constructions under a tightly controlled protocol. This framing is consistent with recent work arguing that empirical conclusions in QML are often regime-dependent and highly sensitive to protocol choices [Schuld et al. \(2021\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#).

3.6 Model selection and repeated-run protocol

A major source of ambiguity in robustness studies is the selection criterion. Strong ID performance does not necessarily imply strong OOD performance, and mixing the two

objectives can make results difficult to interpret. We therefore use two complementary family-internal selection modes.

Best-by-OOD.

Within each family (classical or quantum), we select the configuration with the highest mean OOD balanced accuracy. We then compare the selected classical model against the selected quantum model. This is the robustness-oriented view and constitutes the primary comparison in the paper.

Best-by-ID.

Within each family, we instead select the configuration with the highest mean ID balanced accuracy and then compare the selected classical and quantum models. This is the separability-oriented view and is reported separately to avoid conflating ID accuracy with OOD robustness.

This separation is methodological, not cosmetic. It acts as a guardrail against over-interpreting ID gains as robustness gains and helps disentangle two distinct questions: which kernel family yields the strongest in-distribution class separation, and which yields the strongest out-of-distribution generalization.

Selection is only one source of ambiguity in robustness studies. A second one is result stability: even under fixed setting definitions, conclusions based on single runs can be misleading if variability across split and model seeds is not explicitly accounted for.

To quantify variability, each configuration is repeated across 15 runs. At the orchestration level, this repeated-run protocol is generated by combining multiple split seeds and model seeds under fixed setting definitions, after which results are summarized by mean and standard deviation. This repeated-run structure is important because the paper aims to support variability-aware conclusions rather than single-run anecdotes.

In the implementation used for the paper, repeated-run variability is induced by combining five q-split seeds with three model seeds, yielding 15 runs per evaluated configuration under fixed setting definitions. The master setting itself remains defined by the shift variant, master seed, and size regime; within that setting, both classical and quantum candidates are evaluated on identical split definitions and summarized by mean $\pm$ standard deviation across the repeated runs. This design makes the reported comparisons variability-aware while preserving strict comparability between kernel families.

Statistical testing.

Family-level comparisons are tested formally with paired two-sided Wilcoxon signed-rank tests over the per-setting deltas (the settings are the paired units), separately for each classifier and selection view, with Holm correction applied within each classifier across the reported comparisons. We complement these tests with the per-setting effect-size proxy defined below, so that both the consistency of the sign pattern and the magnitude relative to run-to-run noise are visible.

3.7 Evaluation metrics

Our primary metric is balanced accuracy, reported separately on ID and OOD test sets:

$$\text{BAcc}_{\text{ID}}, \quad \text{BAcc}_{\text{OOD}}.$$

Balanced accuracy is preferred because it remains informative under moderate class imbalance and is straightforward to interpret in both settings.

We quantify robustness under shift through the ID-to-OOD degradation

$$\text{drop} = \text{BAcc}_{\text{ID}} - \text{BAcc}_{\text{OOD}}.$$

Lower values indicate greater robustness, as they correspond to smaller degradation when moving from ID to OOD evaluation.

To compare families after selection, we compute family deltas:

$$\Delta_{\text{OOD}} = \text{BAcc}_{\text{OOD}}^{(Q)} - \text{BAcc}_{\text{OOD}}^{(C)}, \quad (3)$$

$$\Delta_{\text{ID}} = \text{BAcc}_{\text{ID}}^{(Q)} - \text{BAcc}_{\text{ID}}^{(C)}. \quad (4)$$

Positive values indicate an advantage for the selected quantum model relative to the selected classical model.

Because average gains can be misleading when variability is large, we also report an effect-size-to-variability proxy:

$$\text{effect} = \frac{\mu_Q - \mu_C}{\sqrt{\sigma_Q^2 + \sigma_C^2}},$$

where μ_Q, μ_C and σ_Q, σ_C are the repeated-run means and standard deviations for the metric of interest. Values greater than 1 indicate that the observed mean gain exceeds the combined variability scale, which provides a simple and interpretable robustness check on whether an apparent gain is large relative to run-to-run noise.

Probabilistic metrics.

For the Gaussian process classifier we additionally report, on both ID and OOD splits, the negative log-likelihood (log-loss), the Brier score, the expected calibration error (ECE, 15 equal-width bins), and the mean predictive entropy. These quantities characterize how the quality and honesty of predictive uncertainty change under shift: a well-behaved probabilistic model should become *more* uncertain on drifted inputs (rising entropy) while its calibration degrades as little as possible (bounded ECE increase).

3.8 Kernel-geometry descriptors

To move from measuring outcomes to explaining them, we compute geometry descriptors of the Gram matrices that every kernel induces in every setting and dimension. All descriptors are computed from the very kernel matrices consumed by the classifiers (unit diagonal by construction), so the analysis describes exactly the objects the models see.

Effective rank.

For the training Gram matrix K_{tr} with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ normalized to $p_i = \lambda_i / \sum_j \lambda_j$, the effective rank is $\exp(-\sum_i p_i \log p_i)$ Roy and Vetterli (2007). It measures how many directions of feature space the kernel actually uses: a cosine-normalized linear kernel on low-dimensional embeddings is near rank one, while fidelity kernels can spread mass over many directions.

Centered kernel-target alignment.

For a kernel block K on a labeled split with labels $y \in \{-1, +1\}^n$, the centered kernel-target alignment (KTA) is $\langle K_c, yy^\top \rangle_F / (n \|K_c\|_F)$, where K_c is the doubly centered kernel Cristianini et al. (2001); Cortes et al. (2012). We evaluate KTA on the training, ID, and OOD splits separately; the ID-to-OOD change in alignment measures whether the label structure remains visible to the kernel’s geometry after shift—the quantity we will call *alignment survival*.

Concentration and cross-split similarity.

We record the mean and standard deviation of off-diagonal kernel entries within the training block and of the cross blocks between each test split and the training set, together with the ratio of mean OOD-to-train and ID-to-train similarities, which measures how novel the OOD inputs look under each kernel’s similarity notion.

Geometric difference.

Between each classical training kernel K_C and each quantum training kernel K_Q we compute the regularized geometric difference $g(K_C \rightarrow K_Q) = \|\sqrt{K_Q}(K_C + \lambda nI)^{-1}\sqrt{K_Q}\|_\infty^{1/2}$ of Huang et al. [Huang et al. \(2021\)](#), over a sweep of relative regularizers λ . Large values indicate that the quantum geometry can deviate substantially from what the classical kernel expresses; the quantity bounds the *potential* for a performance separation but not its sign.

Mechanistic analyses.

These descriptors feed two complementary analyses. First, a diagnostic analysis at the setting level asks which descriptors of the Best-by-OOD-selected kernels separate classical-favorable from quantum-favorable settings (Mann–Whitney AUC over settings) and correlate with the observed OOD delta. Second, a within-setting analysis asks whether geometry predicts performance across *all* kernel–dimension cells of a setting: for every setting, classifier, and q-split seed we compute the Spearman rank correlation between a descriptor (effective rank, OOD alignment) and OOD balanced accuracy over the 30–55 evaluated cells, and then examine the distribution of these correlations across settings and datasets. A descriptor that yields consistently positive within-setting correlations across datasets, drift mechanisms, and classifiers constitutes a dataset-general regularity rather than a benchmark artifact.

3.9 Experimental grid and reporting

The EMBER evaluation grid comprises 18 principal settings formed by the Cartesian product of:

- two shift variants ($m1, m2$),
- three master seeds,
- and three training-size regimes.

The three size regimes are:

$$(1000, 500, 500), \quad (2000, 1000, 1000), \quad (4000, 1800, 1800),$$

corresponding to $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$.

The network-flow grid applies the same structure to the three scenarios (UNSW-DoS, UNSW-Reconnaissance, ToN-IoT-Scanning), each under two shift mechanisms (m2-centroid and natural drift), three master seeds, and the same three size regimes, yielding 54 principal settings. Together the study comprises 72 principal settings, each evaluated across the dimension sweep and repeated over 15 runs, for both classifiers. The bandwidth-symmetric sweep enlarges the kernel candidate sets on the full EMBER grid and on the natural-drift network settings.

For clarity and reproducibility, Table 1 summarizes the main experimental protocol used throughout the paper, including the shift constructions, controlled seed structure, size regimes, dimensionality sweep, kernel families, classifiers, and repeated-run design.

Within each setting, we evaluate both kernel families across the shared preprocessing grid and the dimension sweep $d \in \{4, 6, 8, 10, 12\}$. This produces both a setting-level comparison, used for the main Best-by-OOD and Best-by-ID analyses, and a broader dimension-level view used to characterize how representation size interacts with kernel-family behavior.

For reproducibility, split definitions are fixed within each setting, random seeds are controlled, and all key quantities are reported as mean $\pm$ standard deviation across repeated runs. This summary makes explicit that fairness in the study is defined at the protocol level: both kernel families are evaluated under the same classifier, preprocessing, split structure, dimensionality sweep, bandwidth-tuning freedom, and repeated-run design, while differing only in the kernel construction itself. Throughout the paper, we deliberately avoid claims of universal dominance or complexity-theoretic

Table 1 Compact summary of the controlled experimental protocol. The repeated-run design is generated by combining five split seeds with three model seeds, yielding 15 runs per evaluated configuration under fixed setting definitions.

Component	Specification
Datasets	EMBER; UNSW-NB15 (DoS, Reconnaissance); ToN-IoT (Scanning)
Shift constructions	EMBER: $m1$, $m2$; network: m2-centroid, natural drift
Master seeds	42, 123, 999
Split seeds	42, 123, 999, 7, 2024
Model seeds	42, 123, 999
Runs per configuration	$5 \times 3 = 15$
Size regimes ($n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}}$)	(1000, 500, 500), (2000, 1000, 1000), (4000, 1800, 1800)
Dimensions	$d \in \{4, 6, 8, 10, 12\}$
Classifiers	SVC (precomputed); Laplace-GP classifier (precomputed)
Shared preprocessing	MaxAbsScaler + TruncatedSVD + angle mapping
Classical-original family	Linear, RBF(scale)
Classical-extended family	+ poly(2,3), Laplacian, Matérn(3/2, 5/2), median heuristic
Quantum family	ZZ (1 rep), ZZ (2 reps), PauliXZ (1 rep), Z-map
Bandwidth sweep	RBF γ factors $\{0.1, 0.3, 1, 3, 10\}$; angle scales $\{0.5, 1, 2\}$
Selection views	Best-by-OOD, Best-by-ID (family-internal, on run means)
Statistical testing	Paired Wilcoxon over settings + Holm; effect proxy
Principal settings	EMBER $2 \times 3 \times 3 = 18$; network $3 \times 2 \times 3 \times 3 = 54$; total 72

quantum advantage. The object of study is narrower and more empirical: under a fixed classifier and a fixed evaluation protocol, does changing only the kernel alter robustness under distribution shift in security-relevant tasks—and which property of the kernel is responsible?

4 Results

4.1 Overview

We report results under the controlled kernel-swap protocol described in Section 3. All comparisons share the same classifier family, preprocessing pipeline, and ID/OOD split definitions; the only varying component is the kernel. Performance is summarized using balanced accuracy on the ID and OOD splits, the ID-to-OOD degradation (*drop*),

and the variability-aware effect proxy defined in Section 3. To keep the empirical questions distinct, we organize the results around two complementary selection modes: *Best-by-OOD*, which targets robustness under shift, and *Best-by-ID*, which targets in-distribution separability.

Our main evaluation spans 18 principal settings obtained from the Cartesian product of the two shift variants, three master seeds, and three training-size regimes. We then complement this setting-level analysis with a finer-grained dimension-level study over $d \in \{4, 6, 8, 10, 12\}$ in order to determine whether the observed trends are stable across representation sizes or concentrated in narrow regions of the grid.

4.2 Best-by-OOD: quantum kernels improve OOD performance in most settings

We begin with the robustness-oriented view. For each setting, we select the classical configuration with the highest mean OOD balanced accuracy and the quantum configuration with the highest mean OOD balanced accuracy, and then compare the two selected models directly. This comparison answers the primary practical question of the paper: if each family is allowed to present its strongest OOD candidate under the same protocol, which family tends to generalize better under shift?

The answer is favorable to quantum kernels in the majority of cases. Across the 18 main settings, the selected quantum model achieves higher OOD balanced accuracy in 14 settings, while the selected classical model is superior in the remaining 4. The mean improvement is

$$\Delta_{\text{OOD}} = +0.0189,$$

and the median improvement is

$$\text{median}(\Delta_{\text{OOD}}) = +0.0255.$$

Table 2 Headline results under the controlled kernel-swap protocol. Best-by-
OOD selects, within each family, the configuration with the highest mean OOD
balanced accuracy. Best-by-ID applies the same logic using ID balanced accuracy.

Selection view	Q wins	C wins	Mean Δ	Median Δ	Effect > 1
Best-by-OOD	14/18	4/18	+0.0189	+0.0255	9/18
Best-by-ID	18/18	0/18	+0.0737	+0.0788	18/18

Moreover, in 9 of the 18 settings, the OOD effect proxy exceeds 1, indicating that the observed mean gain is larger than the combined run-to-run variability scale. This is important because it shows that the overall trend is not driven only by marginal wins: in a substantial subset of settings, the quantum advantage is also large relative to the observed noise level.

These results support the central empirical claim of the paper. Under a strictly controlled kernel-swap comparison, fidelity-based quantum kernels can yield measurable and often variability-dominating improvements in OOD balanced accuracy in malware detection under distribution shift. At the same time, the win count is not uniform enough to justify any claim of universal dominance. The more appropriate interpretation is that quantum kernels are advantageous in most, but not all, of the evaluated regimes.

Figure 1 summarizes the setting-level Best-by-OOD comparison across the 18 principal settings. The win map makes the global pattern immediately visible: quantum kernels dominate in most regimes, while the classical wins are concentrated in a small subset of structured settings rather than being randomly scattered across the grid. Table 2 summarizes the main outcomes under both selection views, while the full setting-level Best-by-OOD comparisons are reported in Appendix A.1.

4.3 Best-by-ID: consistent gains in in-distribution separability

We next turn to the separability-oriented view. For each setting, we now select the best classical configuration by mean ID balanced accuracy and the best quantum

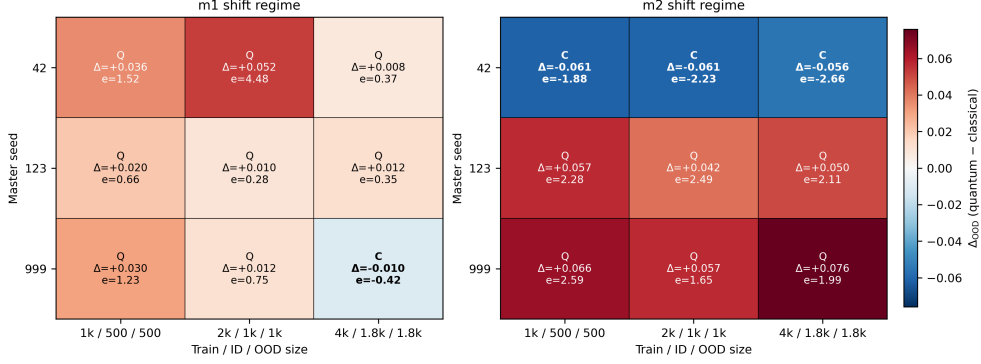


Fig. 1 Best-by-OOD win map across the 18 principal settings. Each cell indicates the winning family and the corresponding OOD gain. Quantum kernels win in 14 of the 18 settings, while the classical-dominant cases are concentrated in a small subset of structured regimes.

configuration by mean ID balanced accuracy, and compare the selected models under ID evaluation. This comparison is not a substitute for the OOD analysis, but it is still scientifically informative because it reveals whether one kernel family more consistently induces favorable decision geometry under the i.i.d. assumption.

Under this criterion, the pattern is strikingly consistent. The selected quantum model achieves higher ID balanced accuracy in all 18 main settings. The mean improvement is

$$\Delta_{\text{ID}} = +0.0737,$$

and the median improvement is

$$\text{median}(\Delta_{\text{ID}}) = +0.0788.$$

The variability-aware signal is also strong: the gains are not only universal across settings, but also systematically large relative to run-to-run noise.

This result suggests that, in the shared reduced feature space used here, the quantum feature maps frequently induce a similarity structure that is more favorable to class separation for SVC than the strongest selected classical alternatives. However,

this should not be misread as implying universal OOD superiority. One of the main reasons for reporting Best-by-ID separately is precisely to avoid conflating stronger ID separation with stronger robustness under shift. The contrast between the 18/18 ID wins and the more qualified 14/18 OOD wins makes that distinction explicit.

The Best-by-ID outcome is summarized in Table 2, while the full setting-level Best-by-ID comparisons are reported in Appendix A.2. This makes the 18/18 ID advantage of the selected quantum models fully transparent and reproducible without overloading the main manuscript.

4.4 Classical-dominant regimes are structured rather than random

Although the Best-by-OOO results favor quantum kernels overall, the exceptions are scientifically important. A strong paper should not hide the four settings where classical kernels win; it should explain why their existence strengthens rather than weakens the study.

The key point is that these exceptions are structured. They are not scattered arbitrarily across the grid, but concentrate in specific regimes, particularly in configurations associated with the $m2$ shift mechanism and, more specifically, with the master seed 42 under the histogram+byte-entropy feature block. This concentration suggests that the outcome depends on the interaction between three factors: the shift construction, the induced feature geometry, and the kernel family’s similarity bias. In other words, the comparison is not revealing a binary hierarchy in which one family is always better; it is revealing a regime-dependent landscape.

This regime-level structure matters for two reasons. First, it prevents over-generalization from the aggregate win counts. A result such as 14/18 is more informative when the 4 losses are interpretable than when they are dismissed as noise.

Table 3 Classical-dominant settings under Best-by-OOD selection. $S = (1k, 500, 500)$, $M = (2k, 1k, 1k)$, and $L = (4k, 1.8k, 1.8k)$. OOD_C and OOD_Q denote mean $\pm$ std balanced accuracy. $Effect_{OOD} = \Delta_{OOD} / \sqrt{\sigma_C^2 + \sigma_Q^2}$.

Var.	Seed	Size	Classical	Quantum	OOD_C	OOD_Q	Δ_{OOD}	$Effect_{OOD}$
m1	999	L	RBF	ZZ-r1 (d=10)	0.763 ± 0.024	0.753 ± 0.005	-0.010	-0.41
m2	42	S	RBF	PauliXZ-r1 (d=8)	0.735 ± 0.013	0.674 ± 0.030	-0.061	-1.87
m2	42	M	Linear	PauliXZ-r1 (d=12)	0.769 ± 0.019	0.708 ± 0.019	-0.061	-2.27
m2	42	L	Linear	PauliXZ-r1 (d=10)	0.784 ± 0.012	0.728 ± 0.017	-0.056	-2.69

Second, it produces actionable guidance: practitioners should not treat quantum kernels as an automatic replacement for classical ones, but as a strong candidate family whose utility depends on the expected shift regime.

Table 3 reports the selected winning configurations in the classical-dominant settings, including the kernel type, embedding dimension, and the corresponding OOD mean $\pm$ std for both families. This makes the “where it fails” story concrete and reproducible rather than anecdotal.

4.5 ID-to-OOD degradation should be read as a complementary robustness signal

In addition to absolute ID and OOD balanced accuracy, we report the degradation

$$drop = BAcc_{ID} - BAcc_{OOD}.$$

This quantity captures how much performance changes when moving from ID to OOD evaluation, and is therefore a complementary indicator of robustness. However, it should not be interpreted in isolation from the absolute OOD scores. In the present study, quantum kernels often improve ID performance very strongly and improve OOD performance in the majority of settings, which means that the *drop* can reflect a

combination of two effects: improved OOD behavior and even larger shifts in the corresponding ID baseline.

Because drop conflates OOD behavior with the corresponding ID baseline, we treat it as a secondary robustness descriptor rather than as the primary ranking criterion. The main inferential target of the paper is OOD balanced accuracy under controlled shift-aware evaluation. Its main role in the paper is interpretive: it helps distinguish settings where OOD gains come with similar ID behavior from settings where one family changes the entire ID/OOD profile more substantially. Read this way, *drop* enriches the regime analysis rather than replacing the OOD comparison.

A setting-level summary of drop can still be useful as supplementary descriptive material, but the main narrative remains anchored on OOD balanced accuracy, since robustness under shift is the primary target of the study.

4.6 Dimension-level analysis

The 18 setting-level comparisons are intentionally complemented by a finer-grained analysis over the dimensionality parameter $d \in \{4, 6, 8, 10, 12\}$. This dimension sweep serves two purposes. First, it tests whether the observed OOD gains are stable across representation sizes rather than artifacts of a single projection dimension. Second, it helps identify whether classical-dominant cases arise from a broad regime or from a narrow band of dimensions.

A consistent pattern emerges from this analysis. In many settings, quantum kernels remain competitive or improve across the dimension sweep, while classical kernels can exhibit either saturation or greater regime sensitivity depending on the shift construction. This dimension-level view also clarifies the classical-dominant settings: some exceptions are not full-setting reversals across all dimensions, but rather regimes in which classical kernels hold an advantage over a wider or more influential subset of

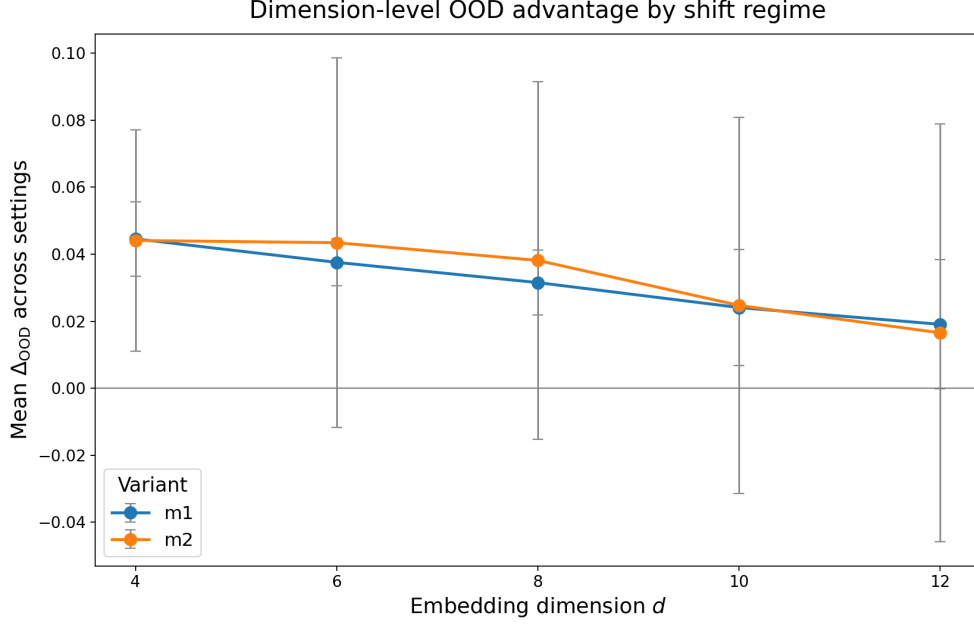


Fig. 2 Dimension-level OOD comparison across $d \in \{4, 6, 8, 10, 12\}$. The plot summarizes how the mean OOD advantage varies with representation size and helps identify whether classical-dominant cases are broad reversals or narrow dimensionality bands.

the dimensionality grid. This is exactly the kind of fine-grained behavior that would be missed by reporting only a single best configuration per family.

Figure 2 shows the dimension-level OOD comparison across $d \in \{4, 6, 8, 10, 12\}$. The figure helps determine whether the observed advantages persist across representation sizes and clarifies whether classical-dominant settings reflect broad reversals or only narrow dimensionality bands. Table 4 summarizes the dimension-level comparison over all 90 setting–dimension cells.

Table 4 confirms that this behavior is not restricted to isolated best-of-grid selections: the quantum family is favorable in 78 of the 90 setting–dimension cells, supporting a broader regime-level effect rather than a narrow configuration artifact.

Table 4 Summary of the dimension-level OOD comparison across the 90 setting–dimension cells.

Statistic	Value
Quantum OOD wins	78/90
Classical OOD wins	12/90
Mean Δ_{OOD}	+0.0324
Median Δ_{OOD}	+0.0383
Effect > 1	56/90

4.7 Summary of empirical findings

Taken together, the empirical results support three main conclusions. First, under robustness-oriented selection, quantum kernels improve OOD balanced accuracy in the majority of evaluated settings, with positive average and median gains and a substantial subset of settings in which the gains exceed the observed variability scale. Second, under ID-oriented selection, quantum kernels improve in-distribution balanced accuracy consistently across all main settings, indicating that the induced similarity geometry is often favorable for class separation in the shared representation space. Third, the exceptions are structured rather than random, which enables a regime-based understanding of when classical kernels remain competitive or preferable.

This is the most defensible reading of the evidence. The results do not justify a claim of universal superiority, but they do justify a stronger and more precise statement: under a fair and reproducible kernel-swap protocol, quantum kernels emerge as a competitive and often advantageous family for malware detection under distribution shift, while still exhibiting identifiable failure regimes that must be part of any honest interpretation.

5 Discussion

5.1 What do the results say about robustness under shift?

The primary objective of this study is robustness under distribution shift rather than maximum in-distribution accuracy. Under the controlled kernel-swap protocol, quantum kernels improve OOD balanced accuracy in the majority of the evaluated settings under Best-by-OOD selection. This finding is meaningful for two reasons. First, the comparison is intentionally strict: the classifier family, preprocessing pipeline, split definitions, and model-selection logic are fixed, so the observed differences are interpretable as differences in kernel behavior rather than as artifacts of heterogeneous pipelines. Second, the gains are not supported only by win counts; several settings also exhibit improvements that exceed the combined run-to-run variability scale, indicating that at least part of the observed advantage is large relative to the experimental noise.

From the perspective of the distribution-shift literature, this reinforces a general lesson: conclusions about robustness are only as credible as the protocol used to obtain them [Quiñonero-Candela et al. \(2008\)](#); [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#). In our case, the relevant conclusion is not that quantum kernels are categorically superior, but that under a fair, explicitly shift-aware comparison, they emerge as a competitive and often advantageous kernel family for malware detection.

At the same time, the evidence should be interpreted as *regime-conditional*. Our OOD splits are induced by two controlled shift mechanisms, not by an exhaustive model of all real malware drift processes. Accordingly, the strongest defensible reading is that quantum kernels improve robustness in many of the shift regimes studied here, not that they would necessarily do so under every temporal, adversarial, or cross-domain drift scenario encountered in practice.

5.2 Consistent ID gains do not imply universal OOD dominance

One of the clearest empirical patterns in the paper is the contrast between the ID and OOD views. Under Best-by-ID selection, quantum kernels improve ID balanced accuracy in all 18 main settings. This is a strong and highly consistent result. A natural interpretation is that, within the shared reduced representation used here, the quantum feature maps often induce a similarity geometry that is favorable for class separation under SVC. This interpretation is compatible with the kernel view of quantum models, in which the feature map changes the implicit geometry of the classification problem rather than the classifier itself [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#).

However, the OOD results are more qualified than the ID results, and that distinction matters. A model family can induce better margins, cleaner separation, or stronger fit on the ID distribution without being uniformly more robust once the evaluation distribution changes. This is exactly why we separated Best-by-ID and Best-by-OOD selection modes. That separation acts as a methodological guardrail: it prevents strong ID performance from being mistaken for direct evidence of robustness.

This point is broader than the present paper. In domain generalization and OOD evaluation, methods often look impressive under one selection criterion but lose their advantage under another [Gulrajani and Lopez-Paz \(2021\)](#). In malware detection, the same caution is even more important because temporal variation and concept drift can alter the relationship between training geometry and future data [Barbero et al. \(2022\)](#); [Li et al. \(2025\)](#). Our results fit this pattern well: the ID gains are real and strong, but the OOD story is more nuanced, which is exactly why the robustness-oriented view must remain primary.

5.3 Why do classical-dominant regimes appear?

The presence of classical-dominant regimes is not a weakness of the study; it is one of its most informative outcomes. If the results had shown quantum wins everywhere, the paper would be more vulnerable to the suspicion that the benchmark was unintentionally tuned toward one family. Instead, the fact that a small number of structured settings remain favorable to classical kernels strengthens the interpretation that the protocol is actually discriminating between regimes rather than merely amplifying one side.

A plausible explanation is that these exceptions arise from interactions between the shift mechanism and the induced kernel geometry. In the $m2$ construction, OOD samples are selected as within-class extremes of a distance-to-train score anchored on the final training set. Such points may occupy regions where some fidelity-based quantum kernels become overly concentrated, less discriminative, or otherwise poorly matched to the relevant notion of locality. This hypothesis is consistent with the broader quantum-kernel literature, which emphasizes the importance of encoding, kernel concentration, and bandwidth-like effects in determining empirical performance [Schuld et al. \(2021\)](#); [Huang et al. \(2021\)](#); [Gil-Fuster et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#). Classical RBF kernels, by contrast, can behave as strong local similarity measures whose effective geometry may align more naturally with certain outlier-driven or centroid-distance-driven OOD constructions.

The key message is therefore not “quantum versus classical” as a universal hierarchy, but *kernel family versus shift regime*. The exceptions reveal that kernel usefulness depends on how the induced similarity notion interacts with the representation and with the way OOD examples are generated. This regime-aware interpretation is much more valuable than a binary winner-takes-all narrative, because it points toward a more actionable question for future work: which properties of a shift regime predict whether a given kernel family will be favorable?

5.4 Implications for security machine learning practice

From a practical perspective, the paper suggests two takeaways for security ML. The first is substantive: when compared fairly under fixed protocols, quantum kernels can provide meaningful OOD robustness gains in malware detection. This makes them worthy of serious consideration as candidate models in security pipelines, especially when the main concern is not just fitting the present distribution but preserving performance under plausible shift.

The second is methodological. Independently of whether one ultimately prefers a classical or quantum kernel, the protocol used here provides a reusable blueprint for robust evaluation in security settings:

1. keep the classifier and preprocessing fixed,
2. define explicit OOD splits tied to interpretable shift mechanisms,
3. repeat runs and report variability rather than single numbers,
4. and separate model-selection criteria according to the actual objective (robustness vs. separability).

This structure is useful beyond the specific models studied here. Security ML frequently suffers from comparisons in which model class, tuning effort, and evaluation conditions change simultaneously. A controlled kernel-swap design avoids that ambiguity and yields conclusions that are easier to trust and easier to reproduce.

Practitioners should nonetheless resist the temptation to interpret these results as a blanket recommendation to replace classical kernels. The more responsible recommendation is narrower: treat quantum kernels as a strong candidate family whose value should be tested under the expected shift regime, not as a default substitute for well-tuned classical baselines.

5.5 Limitations and threats to validity

Several limitations define the scope of our conclusions.

Shift operationalization.

Our OOD splits are constructed through two controlled mechanisms, $m1$ and $m2$. These mechanisms are reproducible and useful for regime comparisons, but they represent only a subset of possible real-world drift processes. Future work should connect the present protocol to additional forms of shift, including explicit temporal splits, family-based drift, and adversarially induced distribution change.

Kernel-family coverage.

We compare strong classical baselines (linear and RBF) against multiple fidelity-based quantum kernels induced by commonly used feature maps. This is a meaningful and fair comparison, but not an exhaustive one. The quantum-kernel design space is much larger, and classical alternatives beyond linear/RBF could also be relevant. Extending the study to broader kernel families would help determine whether the present trends persist or are specific to the compared set.

Selection over candidate sets.

Both Best-by-OOD and Best-by-ID are selection procedures over a finite candidate set. Although the same logic is applied within each family, the outcome still depends on the candidate space and search granularity. This is one reason why repeated-run reporting is important, but it does not eliminate all forms of selection sensitivity. Future work could strengthen the analysis further through formal statistical testing, nested selection protocols, or robustness analyses over expanded search spaces.

Hardware interpretation.

Our quantum kernels are studied as kernel constructions defined through fidelity-based similarities, not as demonstrations of near-term hardware advantage. This distinction is important. The paper evaluates the geometry induced by the kernels, not the

practical performance of noisy quantum devices. We therefore make no claim about hardware superiority, resource advantage, or quantum speedup.

Generality beyond EMBER.

EMBER is a strong benchmark for static PE malware detection, and the updated benchmark landscape further supports its relevance [Anderson and Roth \(2018\)](#); [Joyce et al. \(2025\)](#). Even so, conclusions drawn from one dataset family should not be assumed to transfer automatically to other security modalities such as dynamic malware analysis, network intrusion data, or broader cyber-defense settings. Replication on additional benchmark families will be important for establishing external validity.

5.6 Conservative interpretation

In light of these limitations and the broader QML benchmarking literature, the appropriate interpretation of the paper is deliberately conservative. We do not claim universal superiority, complexity-theoretic quantum advantage, or immediate hardware relevance. Instead, we provide controlled empirical evidence that quantum kernels can improve OOD robustness in a meaningful fraction of settings relevant to malware detection under shift, while also identifying structured regimes in which classical kernels remain competitive or preferable.

This regime-aware view is, in our view, scientifically stronger than a more ambitious narrative. It is consistent with recent calls for careful benchmarking in QML [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#) and with the broader lesson from distribution-shift research that robustness claims should be narrow enough to be credible and specific enough to be testable. Under that standard, the present study supports a clear conclusion: kernel choice matters under shift, quantum kernels deserve serious consideration in this setting, and their benefits are best understood not as universal truths but as structured advantages that depend on the interaction between representation, similarity, and shift regime.

6 Conclusion

This paper asked whether quantum kernels can improve robustness under distribution shift in malware detection when compared fairly against strong classical kernels. To answer this question, we introduced a controlled kernel-swap protocol in which the classifier family, preprocessing pipeline, and split definitions were held fixed, and only the kernel was changed. This design allowed the comparison to be interpreted as a comparison of kernel behavior rather than as a comparison between heterogeneous model pipelines.

Across 18 principal settings on EMBER malware detection, quantum kernels improved OOD balanced accuracy in the majority of cases under robustness-oriented selection, while improving ID balanced accuracy consistently under ID-oriented selection. At the same time, the results were not uniform enough to justify any claim of universal superiority: a small number of structured and localized regimes remained favorable to classical kernels. This is not a weakness of the study, but one of its main outcomes, because it shows that the practical question is not whether quantum kernels always win, but under which shift regimes and representation conditions they help.

The contribution of the paper is therefore twofold. Empirically, it provides variability-aware evidence that quantum kernels can be a competitive and often advantageous family for malware detection under shift. Methodologically, it provides a reproducible evaluation template for controlled comparison under OOD conditions, separating robustness-oriented and separability-oriented selection views and explicitly reporting variability rather than relying on single-run outcomes.

Our claims are deliberately conservative. We do not claim complexity-theoretic quantum advantage, hardware superiority, or universal dominance over classical kernels. Instead, we show that under a fair and explicitly shift-aware protocol, kernel choice materially affects robustness, and that quantum kernels frequently offer an advantage while still exhibiting identifiable, structured failure regimes. In this sense,

the most defensible interpretation of the evidence is regime-aware rather than universal: the benefits of quantum kernels are best understood as depending on the interaction between similarity geometry, representation, and shift regime.

Several directions follow naturally from this work. Future studies should extend the protocol to additional forms of malware drift, broader classical and quantum kernel families, and further security-relevant datasets. More generally, the results suggest that the most fruitful path forward is not to treat quantum and classical kernels as competing absolutes, but to understand more precisely how kernel geometry interacts with representation and shift regime. The central question is therefore not whether one kernel family universally dominates, but which similarity geometry aligns best with the target shift regime under controlled and reproducible evaluation.

Appendix A Additional Results

This appendix provides the full setting-level tables underlying the main-text summary results. To preserve readability in the main manuscript, the core narrative is supported there by compact summary tables and figures, while the complete per-setting comparisons are reported here for transparency and reproducibility. For compactness, size labels denote $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$ as follows: S = (1k, 500, 500), M = (2k, 1k, 1k), and L = (4k, 1.8k, 1.8k).

A.1 Full Best-by-OOD comparisons

Tables [A1](#) and [A2](#) report the full setting-level Best-by-OOD comparisons, split by shift variant for readability.

A.2 Full Best-by-ID comparisons

Tables [A3](#) and [A4](#) report the full setting-level Best-by-ID comparisons, again split by shift variant for readability.

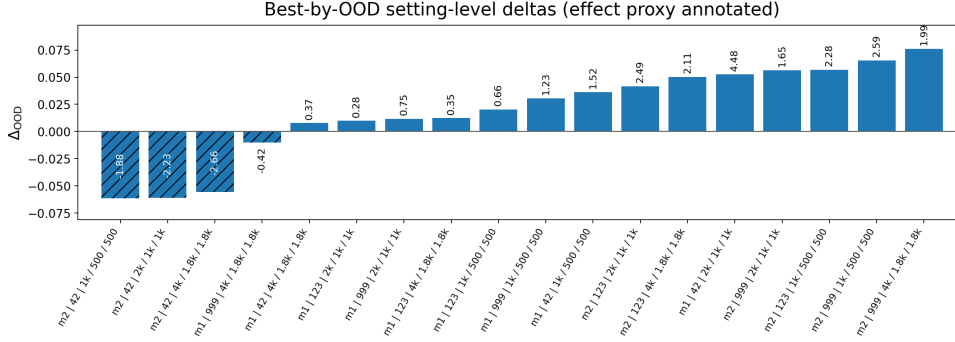


Fig. A1 Setting-level OOD deltas across the 18 principal settings. Positive values indicate an advantage for the selected quantum model over the selected classical model under Best-by-OOO selection.

Table A1 Full Best-by-OOO comparison for shift variant m1.

Var.	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Win.
m1	42	S	RBF(scale)	PauliXZ-r1 (d=12)	0.722±0.013	0.758±0.020	+0.036	Q
m1	42	M	RBF(scale)	ZZ-r1 (d=12)	0.709±0.009	0.761±0.007	+0.052	Q
m1	42	L	RBF(scale)	ZZ-r1 (d=12)	0.749±0.020	0.756±0.008	+0.008	Q
m1	123	S	RBF(scale)	PauliXZ-r1 (d=12)	0.729±0.024	0.750±0.020	+0.020	Q
m1	123	M	RBF(scale)	ZZ-r1 (d=10)	0.746±0.020	0.756±0.029	+0.010	Q
m1	123	L	RBF(scale)	ZZ-r1 (d=10)	0.741±0.032	0.753±0.014	+0.012	Q
m1	999	S	RBF(scale)	PauliXZ-r1 (d=12)	0.737±0.012	0.767±0.022	+0.030	Q
m1	999	M	RBF(scale)	ZZ-r1 (d=12)	0.750±0.009	0.762±0.013	+0.012	Q
m1	999	L	RBF(scale)	ZZ-r1 (d=10)	0.763±0.024	0.753±0.005	-0.010	C

Table A2 Full Best-by-OOD comparison for shift variant m2.

Var.	Seed	Size	Classical	Quantum	OOD C	OOD Q	Δ_{OOD}	Win.
m2	42	S	RBF(scale)	PauliXZ-r1 (d=8)	0.735 ± 0.013	0.674 ± 0.030	-0.061	C
m2	42	M	Linear	PauliXZ-r1 (d=12)	0.769 ± 0.019	0.708 ± 0.019	-0.061	C
m2	42	L	Linear	PauliXZ-r1 (d=10)	0.784 ± 0.012	0.728 ± 0.017	-0.056	C
m2	123	S	RBF(scale)	ZZ-r1 (d=12)	0.729 ± 0.017	0.786 ± 0.018	+0.057	Q
m2	123	M	RBF(scale)	ZZ-r2 (d=12)	0.741 ± 0.005	0.782 ± 0.016	+0.042	Q
m2	123	L	RBF(scale)	ZZ-r2 (d=12)	0.747 ± 0.022	0.797 ± 0.009	+0.050	Q
m2	999	S	RBF(scale)	ZZ-r1 (d=10)	0.709 ± 0.024	0.774 ± 0.007	+0.066	Q
m2	999	M	RBF(scale)	ZZ-r1 (d=12)	0.725 ± 0.030	0.782 ± 0.017	+0.057	Q
m2	999	L	RBF(scale)	ZZ-r1 (d=12)	0.724 ± 0.037	0.800 ± 0.009	+0.076	Q

Table A3 Full Best-by-ID comparison for shift variant m1.

Var.	Seed	Size	Classical	Quantum	ID C	ID Q	Δ_{ID}	Win.
m1	42	S	RBF(scale)	ZZ-r1 (d=12)	0.711 ± 0.024	0.778 ± 0.022	+0.067	Q
m1	42	M	RBF(scale)	ZZ-r1 (d=10)	0.702 ± 0.005	0.793 ± 0.006	+0.090	Q
m1	42	L	RBF(scale)	ZZ-r2 (d=12)	0.719 ± 0.005	0.805 ± 0.008	+0.086	Q
m1	123	S	RBF(scale)	ZZ-r1 (d=12)	0.733 ± 0.020	0.790 ± 0.013	+0.057	Q
m1	123	M	RBF(scale)	ZZ-r1 (d=12)	0.716 ± 0.011	0.798 ± 0.016	+0.082	Q
m1	123	L	RBF(scale)	ZZ-r2 (d=12)	0.712 ± 0.011	0.801 ± 0.006	+0.090	Q
m1	999	S	RBF(scale)	ZZ-r1 (d=12)	0.718 ± 0.030	0.773 ± 0.023	+0.055	Q
m1	999	M	RBF(scale)	ZZ-r2 (d=12)	0.720 ± 0.011	0.793 ± 0.012	+0.073	Q
m1	999	L	RBF(scale)	ZZ-r2 (d=12)	0.722 ± 0.017	0.802 ± 0.006	+0.080	Q

Table A4 Full Best-by-ID comparison for shift variant m2.

Var.	Seed	Size	Classical	Quantum	ID C	ID Q	Δ_{ID}	Win.
m2	42	S	RBF(scale)	ZZ-r1 (d=12)	0.706 ± 0.015	0.772 ± 0.014	+0.066	Q
m2	42	M	RBF(scale)	ZZ-r1 (d=12)	0.759 ± 0.014	0.797 ± 0.014	+0.037	Q
m2	42	L	RBF(scale)	ZZ-r1 (d=12)	0.767 ± 0.012	0.820 ± 0.007	+0.053	Q
m2	123	S	RBF(scale)	ZZ-r1 (d=12)	0.709 ± 0.022	0.784 ± 0.010	+0.076	Q
m2	123	M	RBF(scale)	ZZ-r2 (d=12)	0.714 ± 0.020	0.792 ± 0.013	+0.079	Q
m2	123	L	RBF(scale)	ZZ-r2 (d=12)	0.712 ± 0.007	0.793 ± 0.006	+0.081	Q
m2	999	S	RBF(scale)	ZZ-r2 (d=12)	0.706 ± 0.024	0.798 ± 0.016	+0.092	Q
m2	999	M	RBF(scale)	ZZ-r2 (d=12)	0.702 ± 0.019	0.787 ± 0.009	+0.085	Q
m2	999	L	RBF(scale)	ZZ-r2 (d=12)	0.715 ± 0.019	0.793 ± 0.006	+0.079	Q

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Data Availability. This study is based on the EMBER benchmark dataset [Anderson and Roth \(2018\)](#). The derived split definitions, aggregated results, experiment manifests, and summary tables used in this work are publicly available in the archived project repository: https://github.com/roberto-fernandez-barrios/kernel_shift_framework. Release DOI: <https://doi.org/10.5281/zenodo.19152497>.

Code Availability. The code used to generate the shift splits, create q-splits, run the classical and quantum kernel experiments, aggregate repeated-run results, and reproduce the reported tables and figures is publicly available at https://github.com/roberto-fernandez-barrios/kernel_shift_framework. Archived release DOI: <https://doi.org/10.5281/zenodo.19152497>.

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